

Project Initialization and Planning Phase

Date	03 October 2026
Team ID	SWTID-2026-5999
Project Name	PocketSmart AI — AI-Assisted Goal-Based Budget Planner
Maximum Marks	5 Marks

Project Proposal

PocketSmart AI — AI-Assisted Goal-Based Budget Planner is a web application that converts a user’s spending limit and practical requirements into a structured plan for home setup, event preparation or occasion-based jewelry selection.

Project Overview

Field	Details
Objective	Help a user decide how to distribute a fixed budget before beginning detailed shopping
Scope	Home, Party and Jewelry planners; AI recommendations; server-side budget checking; accounts; history; optional outfit image; external search links
Out of Scope	Payments, direct retailer checkout, guaranteed live prices and training a custom AI model

Problem Statement

Field	Details
Problem	Planning is often done item-by-item, so users discover that the total is too high only after spending time comparing products
Impact	This causes repeated searching, difficult trade-offs and uncertainty about which categories deserve more of the available budget

Proposed Solution

Field	Details
Approach	Collect a total budget and planner-specific inputs, create a structured Gemini request, parse the response, recompute totals in the backend, trim over-budget items when necessary and present the verified plan
Key Features	Budget-first planning; three use cases; optional outfit-image analysis; category allocation; remaining-budget display; shopping search links; account/history; AI fallback; input validation

Resource Requirements

Resource	Requirement
Hardware	Ordinary laptop/desktop with internet access; no local GPU required
Memory	4 GB RAM or more recommended for development
Storage	About 1 GB free for source, dependencies, database and uploads
Software	Python 3.10+, Git, VS Code or another editor, browser
Main Libraries	FastAPI, Uvicorn, Pydantic, google-genai, Pillow, bcrypt, python-jose, python-dotenv, pytest
Data	Runtime user inputs, generated plans and uploaded images; no project-specific model training dataset

Design Principle

AI proposes; application rules decide. This keeps the generative component useful without allowing an AI-generated total to bypass the application’s budget constraint.